Studying the Putnam

W.R. Casper

Problem Solving Seminar

Department of Mathematics
California State University Fullerton

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What is the Putnam Exam?

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Introduction

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- morning session: 8AM-11AM (6 problems)
- afternoon session: 1PM-4PM (6 more problems)

Benefits

Why take the Putnam?

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- departmental scholarships
- recommendation letters
- not just for supergeniuses or the criminally insane

Getting Putnam practice

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- MAA Mathematics Magazine

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- College Math Journal

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- Kiran Kedlaya's website

Putnam 2016 A1:
Find the smallest integer j such that for every polynomial $p(x)$ with integer coefficients

$$p^{(j)}(k) \ \text{ is divisible by 2016 for all integers } k.$$

Strategy

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- try to explore *specific cases*

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Topics: polynomials, divisibility

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- **Fundamental Theorem of Arithmetic**:
  every positive integer may be written as a product of primes

  $$n = p_1 p_2 \dots p_r$$

  moreover, given another prime factorization

  $$n = q_1 q_2 \dots q_s$$

  then $r = s$ and the p_j's and q_j's are the same, after reordering

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Also, polynomials are linear combinations of **monomials**: x^n

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Scratch work

$$
\begin{aligned}
f(x) &= x \\
f'(k) &= 1 \\
f''(k) &= 0 \\
f'''(k) &= 0 \\
f''''(k) &= 0
\end{aligned}
$$

Scratch work

$$
\begin{aligned}
f(x) &= x^2 \\
f'(k) &= 2k \\
f''(k) &= 2 \\
f'''(k) &= 0 \\
f''''(k) &= 0
\end{aligned}
$$

Scratch work

$$
\begin{aligned}
f(x) &= x^3 \\
f'(k) &= 3k^2 \\
f''(k) &= 6k \\
f'''(k) &= 6 \\
f''''(k) &= 0
\end{aligned}
$$

Scratch work

$$f(x) = x^4$$
$$f'(k) = 4k^3$$
$$f''(k) = 12k^2$$
$$f'''(k) = 24k$$
$$f''''(k) = 24$$

Scratch work

$$f(x) = x^{29}$$
$$f'(k) = 29k^{28}$$
$$f''(k) = 29 \cdot 28k^{27}$$
$$f'''(k) = 29 \cdot 28 \cdot 27k^{26}$$
$$f''''(k) = 29 \cdot 28 \cdot 27 \cdot 26k^{25}$$

How long until divisible by 2016?

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We need five 2's, two 3's and a 7:

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Go one more to get a 2:

$$f^{(6)}(k) = 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25 \cdot 24k^{23}$$

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$$f^{(7)}(k) = 27 \cdot 26 \cdot 25 \cdot 24 \cdot 23 \cdot 22 \cdot 21k^{20}$$

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- Break up your solution into manageable pieces with Lemmas
- Double-check you answered the original question!

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Now, for $f(x) = x^{55}$ we can calculate

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which is not divisible by 2016, since it isn't divisible by 2^5.

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Therefore the answer is at least 8.
To prove that 8 is enough, we'll prove two Lemmas.

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Lemma: If n and k are integers with $k > 0$, then one of the integers
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must be a multiple of k. **Proof**:

The value of n modulo k, ie. the remainder of n/k, is an integer r with $0 \leq r \leq n-1$.

Therefore $n-r$ will be 0 modulo k, ie. it will be a multiple of k.

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$f^{(8)}(x) = n(n-1)(n-2)(n-3)(n-4)(n-5)(n-6)(n-7)x^{n-8}$,

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$N = n(n-1)(n-2)(n-3)(n-4)(n-5)(n-6)(n-7)$.

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$N = n(n-1)(n-2)(n-3)(n-4)(n-5)(n-6)(n-7)$. Consider the numbers $\{n, n-1, n-2, n-3, n-4, n-5, n-6, n-7\}$.

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$N = n(n-1)(n-2)(n-3)(n-4)(n-5)(n-6)(n-7)$. Consider the numbers $\{n, n-1, n-2, n-3, n-4, n-5, n-6, n-7\}$. At least four of the above numbers must be even, and therefore divisible by 2. Actually, one must be a multiple of 4! Likewise, at least 2 will be multiples of 3. Finally, one of the numbers will be a multiple of 7. Hence N and $f^{(j)}(k)$ is divisible by 2016.

Writing up the solution

To finish our solution, note that any polynomial will be of the form

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_n x^n$$

for some integers $a_0, \dots, a_n$.

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By our Lemma, when we take the j'th derivative for $j \geq 8$, each of the powers of x becomes a multiple of 2016.

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This makes $f^{(j)}(k)$ a linear combination of multiples of 2016, and therefore also a multiple of 2016.

Example problem

Try it yourself!:
Find the smallest integer j such that for every polynomial $p(x)$ with integer coefficients

$$p^{(j)}(k) \text{ is divisible by 2024 for all integers } k.$$